

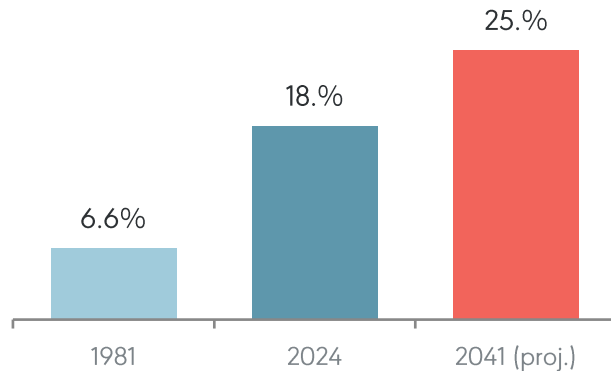


A wearable for elder care homes that alerts carers when a resident tries to stand up alone or needs changing, and records every day to spot decline early. No cameras.

Care demand is growing faster than the care workforce

18%

of Sri Lankans are 60+, up from 6.6% in 1981



1 in 4 will be 60+ by 2041

314,828

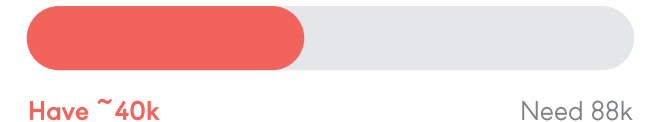
Sri Lankans left for jobs abroad in 2024, a record



Adult children move out and abroad, so parents move into care homes

~40,000

nurses in Sri Lanka, against the 88,000 the WHO minimum calls for



Nurses and carers keep leaving for jobs in Israel, Japan and the Gulf

Care breaks down at night, unseen

81%

of care home falls are unwitnessed

83% happen in the resident's own room, where no one is watching.



Help comes late

In one study, 31% of residents who fell waited over an hour on the floor.



Carers stretched thin

Two or three carers per home, many untrained, covering every room at night.



Nights leave no record

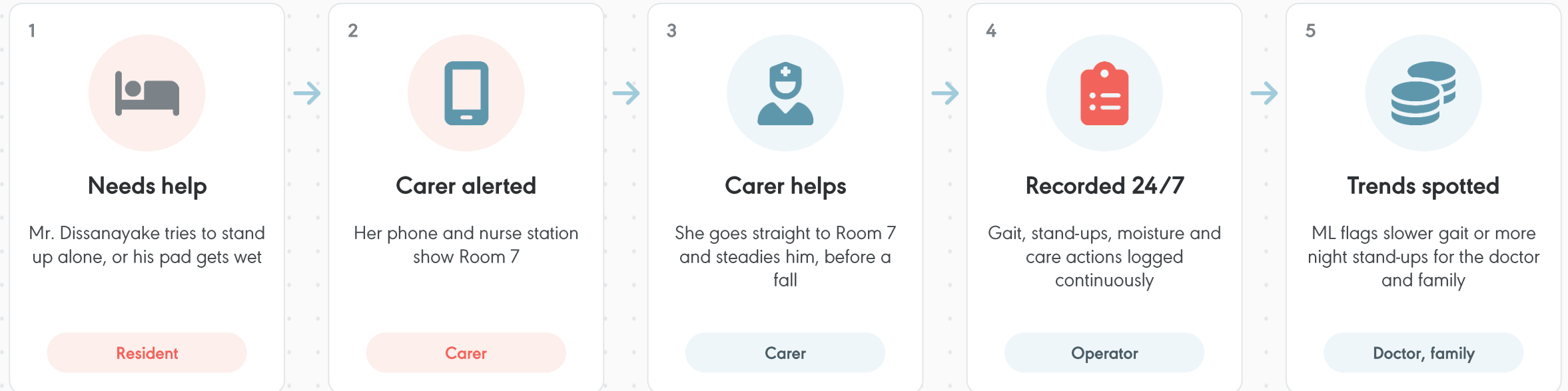
Handover runs on memory. Families get promises, not proof.

"This is not an eight-hour duty. We must work 24 hours."

Carer, aged care home in Colombo

Today's workaround: two-hourly rounds, call bells residents can't press, and CCTV no one watches at 2 a.m.

Help in seconds, a record that builds itself



ALERTS

RECORDS AND TRENDS

TODAY

Nobody knows

Waits for the next round

Checks every room, every two hours

Notes from memory, if at all

Decline noticed only after a fall

The carer always decides. The system detects, alerts, records and spots trends.

Detects on the body, alerts over the LAN



Thigh node

Gait model on the device spots stand-up attempts; moisture and skin temp sensors



Ward gateway

Gets every node over the home's LAN; rule engine checks each bed every 10 s; works without internet



Backend Servers

Stores every reading; finds trends in gait, night stand-ups and moisture over weeks



Apps

Carer phones, nurse station, family app; AI writes the shift handover

97.65%

gait classification accuracy, on the device (lab data)

Seconds

from a stand-up attempt to the carer's phone, no internet needed

24/7

continuous recording. No cameras, ever

ROADMAP

- Now: TRL 5, working prototype
- Months 1 to 8: full build
- Months 9 to 10: testing, then release

Three shifts opened the window in 2026



Edge AI got cheap

THEN

Spotting falls needed cameras or a constant cloud link.

NOW

A \$219 (LKR 72,700) gateway runs the gait model for a whole ward, offline, with \$1.39 (LKR 460) motion sensors on each resident.



Regulation is coming

JUNE 2026

A fire at an unregistered home in Horana killed 12 residents.

NOW

Only 148 of 455 homes are registered. Stricter rules are being drafted.



Privacy became law

THEN

No rules on health data. Cameras in bedrooms went unquestioned.

NOW

Data protection law treats health data as sensitive. Camera-free wins.

Start in Sri Lanka, built for Asia's care homes

TAM · paying beds in Sri Lanka, Malaysia, India

~13,900 beds

\$2.9M (LKR 950M) / yr
+ \$0.76M (LKR 253M) hardware

By 2030: ~22,800 beds, \$4.5M (LKR 1.5B) a year, with India at its projected need

SAM · Sri Lanka's paying segment

~2,150 beds

\$0.18M (LKR 60M) / yr
+ \$0.12M (LKR 39M) hardware

SOM by year 3 · ~10 sites

~300 beds

\$25k (LKR 8.4M) / yr
+ \$16.5k (LKR 5.5M) hardware

THE FORMULA

**Beds × \$7 (LKR 2,300) a month × 12 +
\$55 (LKR 18,400) hardware per bed,
once**



FIRST TARGET

**Registered, fee-paying care homes in
Western Province**

NEXT

Care homes across Asia facing the same ageing
and staff gaps, then hospital and step-down
wards.

One wearable for alerts, moisture and trends

	Bed-exit sensors (Alert-iT, DAZA)	TENA SmartCare Change Indicator	Camera AI (Kepler Night Nurse)	Radar (Vayyar Care)	SoterCare
Alerts when a resident tries to stand up alone	✓	✗	✓	⦿	✓
Detects a wet pad	✗	✓	✗	✗	✓
Records gait and activity continuously	✗	⦿	⦿	⦿	✓
ML trends over weeks, per resident	✗	✗	⦿	⦿	✓
No cameras in the bedroom	✓	✓	✗	✓	✓
Keeps alerting without internet	✓	⦿	✓	⦿	✓
Typical price	€170 to €720 per bed	Sensor plus strips every 12 weeks	Ceiling camera per room, enterprise	~\$240 per device + monthly fee	\$55 (LKR 18,400) per bed + \$7 (LKR 2,300) a month

⦿ Partly: TENA logs pad saturation only; camera AI and radar log room activity, not per-resident gait; cloud-linked alerts depend on the network.

Moat: the only single device doing all six, plus a growing per-resident gait dataset our trend models learn from.

Healthy hardware margins, then a fee per bed

THIGH NODE

\$35 (LKR 11,600)

one per resident

Our cost \$19 (LKR 6,300)

Gross margin **46%**

WARD GATEWAY

\$350 (LKR 116,200)

one per ward, up to ~20 beds

Our cost \$219 (LKR 72,700)

Gross margin **37%**

SUBSCRIPTION

\$7 (LKR 2,300)

per bed per month

Cost to serve ~\$2 (LKR 660)

Gross margin **~70%**

Per bed: ~\$55 (LKR 18,400) hardware once, ~\$24 (LKR 7,900) of it our margin, then \$84 (LKR 27,900) a year. A 20-bed ward kit sells for \$1,110 (LKR 368,500).

All of it is under **5%** of a typical monthly room fee (LKR 50k to 150k).

What we have been doing!



All in [SoterCare.com/news](https://www.sotercare.com/news)



WORKING WITH



We build the whole stack ourselves



Daham Dissanayake

Founder · IOT and ML

Built the node, gateway and on-device gait model.



Sanjula Herath

Co-founder · Backend and AI

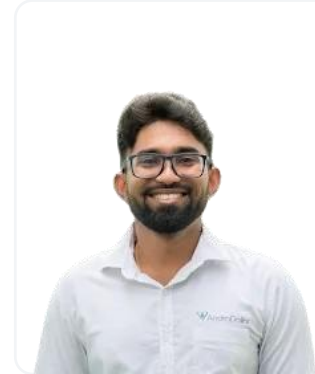
Built the cloud backend, mobile app and AI agents.



Komudi Senarachchi

Co-founder · Design and Docs

Designed the apps, manage all the documentations.



Banu Athuraliya

ADVISOR

Visiting Lecturer/SDGP
Module Leader
Digital Consultant
Informatics Institute of
Technology (IIT)



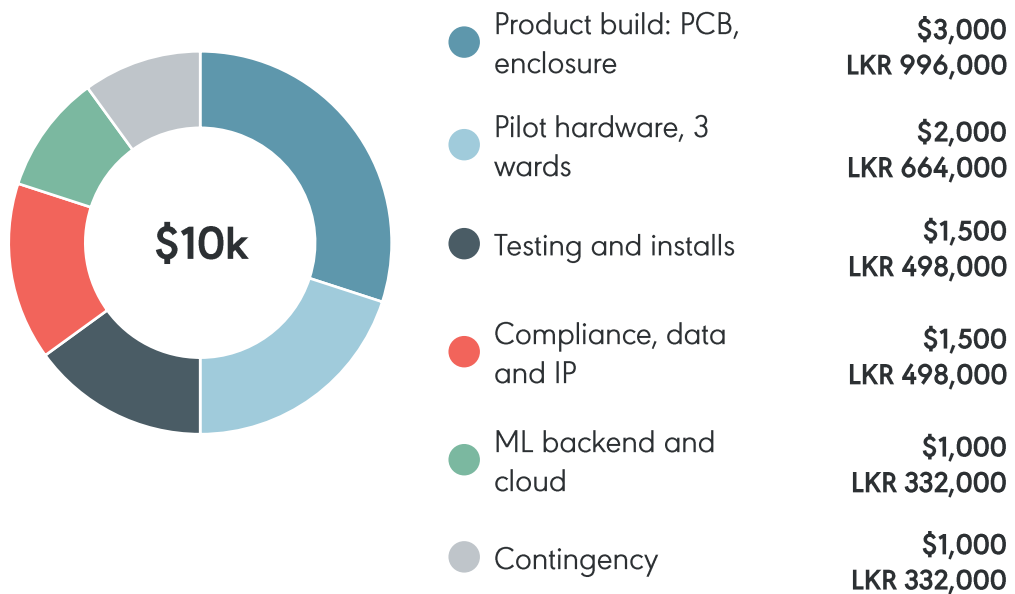
Joining Soon!

MEDICAL EXPERT

Why us: SoterCare started with caring for our own grandparents, and every layer you've seen, from firmware to ML, is ours.

Raising \$10,000 (LKR 3.32M)

USE OF FUNDS · 12 MONTHS



Pilot hardware at ~\$32 (LKR 10,500) a bed. NIA voucher covers part of compliance and the field trial.

Evidence at the next raise: real-world stand-up alerts, false alarms, response times, early trend data, and a paying home.

MILESTONES

Months 1 to 8

Build the full system

Production node and gateway, LAN alerting, moisture sensing, ML trend backend, carer and family apps

Months 9 to 10

Test in 3 homes, 60 beds

Targets: 80% wear time, under 1 false alarm per bed a week, response under 3 minutes

Month 11 onward

Release and first paying homes

Pilot homes convert at a pre-agreed price; outcome report ready for a seed round



VISION

Smarter care for every elder,
As we simplify the wellness!



sotercare.com

+94 70 4888 440

linkedin.com/company/sotercare



Market sizing: the working

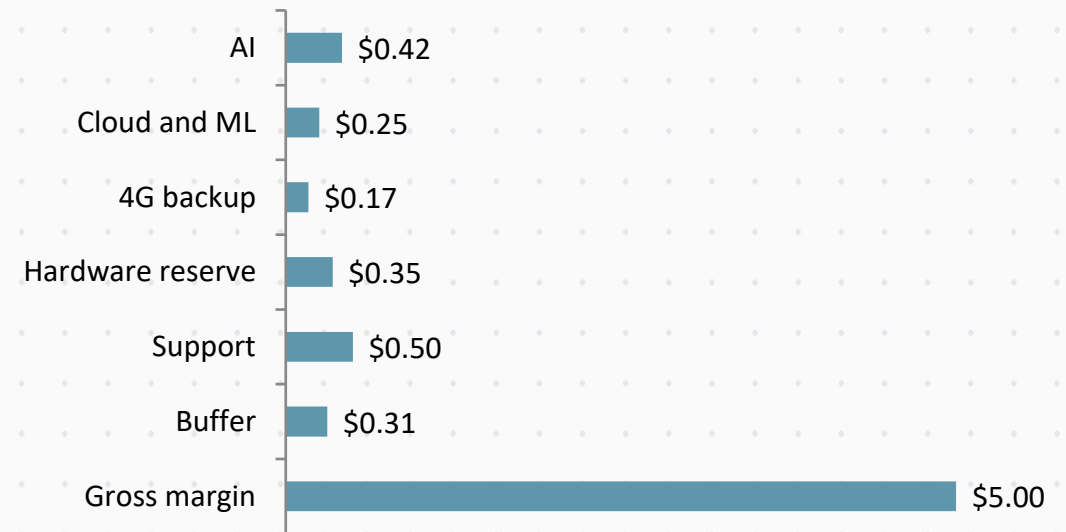
Market and segment	Beds	Fee / bed / month	Subscription / year	Hardware, once
Sri Lanka: fee-paying registered homes (~50 × 25 beds)	1,250	\$7 (LKR 2,300)	\$105k (LKR 34.9M)	\$69k (LKR 22.8M)
Sri Lanka: private hospital long-stay and geriatric wards	800	\$7 (LKR 2,300)	\$67k (LKR 22.3M)	\$44k (LKR 14.6M)
Sri Lanka: paid hospices	100	\$7 (LKR 2,300)	\$8k (LKR 2.8M)	\$5.5k (LKR 1.8M)
SAM: Sri Lanka's paying segment	~2,150		\$0.18M (LKR 60M)	\$0.12M (LKR 39M)
Malaysia: registered centres and licensed nursing homes	~9,600	\$20 (LKR 6,600)	\$2.30M (LKR 765M)	\$0.53M (LKR 175M)
India: organised assisted living, today	~2,100	\$15 (LKR 5,000)	\$0.38M (LKR 125M)	\$0.12M (LKR 38M)
TAM today	~13,900		\$2.9M (LKR 950M)	\$0.76M (LKR 253M)
India: projected need by 2030 (replaces today's 2,100)	~11,000	\$15 (LKR 5,000)	\$1.98M (LKR 657M)	\$0.61M (LKR 201M)
TAM by 2030	~22,800		\$4.5M (LKR 1.5B)	\$1.25M (LKR 415M)
SOM, year 3: ~6 homes, 3 hospital wards, 1 hospice	~300	\$7 (LKR 2,300)	\$25k (LKR 8.4M)	\$16.5k (LKR 5.5M)

USD to LKR at 332. Hardware is priced at ~\$55 (LKR 18,400) per bed in every market. Subscription is set at about 3% of a typical fee in each market.

Hardware pricing, margins and cost to serve

Item	Our cost	Price	Margin
Thigh node	\$19 (LKR 6,300)	\$35 (LKR 11,600)	46%
Ward gateway	\$219 (LKR 72,700)	\$350 (LKR 116,200)	37%
Charging dock (5-bay)	\$9 (LKR 3,100)	\$15 (LKR 5,000)	39%
20-bed ward kit	\$635 (LKR 210,800)	\$1,110 (LKR 368,500)	43%
Per bed	\$32 (LKR 10,500)	\$55 (LKR 18,400)	43%

Where \$7 (LKR 2,300) per bed per month goes

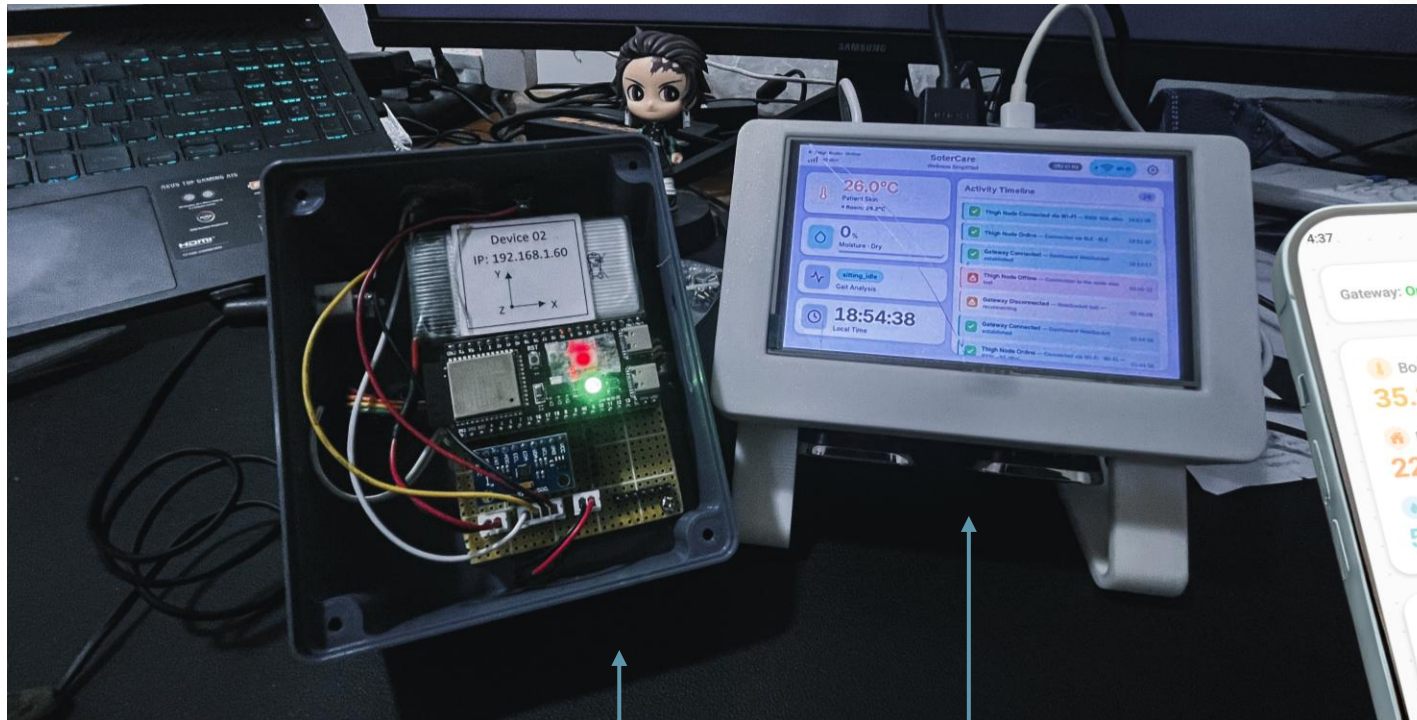


Kit: 20 nodes, 4 docks, 1 gateway. Landed cost at 1,000 nodes and 100 to 500 gateways: distributor price breaks, +3% yield, +15% freight and duty. Excludes one-time enclosure tooling (\$5k to \$12k, LKR 1.66M to LKR 3.98M). Engineering estimates, to be replaced with supplier quotes.

Known risks, and how we handle them

Risk	Mitigation
Residents remove or refuse the wearable, especially with dementia	Soft thigh strap under clothing; measure wear time per resident; target 80%
Charging becomes a burden on scarce staff	7-day battery target; swap-and-charge dock per ward; low-battery alerts
False stand-up alerts cause alarm fatigue	Report false alarms per bed a week, not lab accuracy; carer confirm or dismiss retrains the model
Lab gait accuracy does not transfer to real homes	Testing phase logs every stand-up attempt against every alert
Home Wi-Fi or LAN is unreliable	Site survey before install; gateway buffers data and alerts locally
Medical device classification (NMRA)	Position as safety monitoring and records, not diagnosis; ask NMRA for categorisation early
Homes with low budgets can't pay upfront	Start with fee-paying homes; hardware-as-a-service option; CSR sponsors for charity homes

Current working prototype



Mobile App



Smart Thigh Node (Prototype)

ESP32-S3 · IMU · IR temp · moisture clip
· haptic feedback · SOS



Edge Gateway

Raspberry Pi 5 · local ML · 5"
touchscreen dashboard · voice +
buzzer

Final Product Designs



The Thigh Node



The Ward Gateway

Proactive Gait Analysis Model

A custom on-device classifier that predicts instability before a fall – paired with a private haptic warning.



SoterCare Studio - custom data-capture tool (50 Hz IMU)

97.65%

testing accuracy

Keras NN + K-means anomaly detection,
deployed as an AARCH64 library on the
Pi 5

Five movement states classified in real time

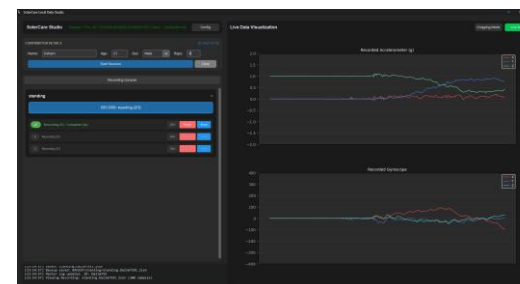
Walking

Standing idle

Sitting idle

Standing up

Sitting down



A Multi-Agent AI Brain

Retrieval-Augmented Generation - three asynchronous pipelines that reason with medical context instead of dumping raw data.



Safety Guard

Real-time · no LLM · Agent 1

Polls vitals every 10 seconds, checks physiological thresholds, writes alerts to the DB, and pushes WebSocket notifications. No LLM.

db_poller · nurse_agent



ARIA Chat bot

Chat · RAG · Agent 2

Provides a RAG chatbot answering questions about the patient and their behavior, using health summaries for Agent 3.

health_agent · FAISS RAG



Clinical Report Agent

6h · persistent · Agent 3

Fetches 6-hour vitals windows, retrieves historical context from Agent 2, and generates structured health reports via LLM.

report_rag · ChromaDB